

Name: _____

Date: _____

LEARNING TARGET

I can solve area, surface-area, and volume problems using formulas and nets.

KEY STRATEGY + WORKED EXAMPLE

Identify the measure requested, label dimensions, choose the matching formula, and track square versus cubic units.

For a 4x3x2 prism, $SA=2(12+8+6)=52$ square units; $V=24$ cubic units.

A. TRY TOGETHER - USE THE STRATEGY

1. Rectangular prism 8 x 5 x 6: surface area = ____ square units
2. Rectangular prism 12 x 7 x 4: surface area = ____ square units
3. Rectangular prism 9 x 9 x 3: surface area = ____ square units
4. Rectangular prism 15 x 6 x 8: surface area = ____ square units

B. CORE PRACTICE - SHOW YOUR WORK

5. Rectangular prism 10 x 4 x 11: surface area = ____ square units
6. Rectangular prism 14 x 5 x 9: surface area = ____ square units

7. Rectangular prism 7 x 13 x 6: surface area = ____ square units

8. Rectangular prism 16 x 8 x 5: surface area = ____ square units

9. Rectangular prism 11 x 10 x 7: surface area = ____ square units

10. Rectangular prism 18 x 6 x 4: surface area = ____ square units

11. Rectangular prism 8 x 5 x 6: surface area = ____ square units

12. Rectangular prism 12 x 7 x 4: surface area = ____ square units

C. INDEPENDENT FLUENCY - CALCULATE AND CHECK

13. Rectangular prism 9 x 9 x 3: surface area = ____ square units
14. Rectangular prism 15 x 6 x 8: surface area = ____ square units
15. Rectangular prism 10 x 4 x 11: surface area = ____ square units
16. Rectangular prism 14 x 5 x 9: surface area = ____ square units
17. Rectangular prism 7 x 13 x 6: surface area = ____ square units
18. Rectangular prism 16 x 8 x 5: surface area = ____ square units
19. Rectangular prism 11 x 10 x 7: surface area = ____ square units
20. Rectangular prism 18 x 6 x 4: surface area = ____ square units

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D. APPLY THE SKILL

21. Find the area of a triangle with base 14 cm and height 9 cm.

22. A box is 10x6x4 inches. Find surface area and volume.

23. A composite figure contains rectangles 8x5 and 3x4 without overlap. Find total area.

E. REASON, CHECK AND CORRECT

24. Explain why surface area and volume use different units and formulas.

25. Correct: Triangle area with $b=12$, $h=7$ is 84 square units.

EXIT TICKET + CHALLENGE

26. A prism has $V=360 \text{ cm}^3$, length=10, width=6. Find height and surface area.

Challenge: Draw a net for a prism, label dimensions, and verify its surface area.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Explain one strategy you used today.

28. Check one answer in a different way.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: _____

31. The most difficult part was: _____

Why was it difficult? _____

32. My next practice step is: _____

TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

- A. 1) 236, 2) 320, 3) 270, 4) 516
B. 5) 388, 6) 482, 7) 422, 8) 496, 9) 514, 10) 408, 11) 236, 12) 320
C. 13) 270, 14) 516, 15) 388, 16) 482, 17) 422, 18) 496, 19) 514, 20) 408
D. 21) 63 square cm. 22) $SA=248$ square in; $V=240$ cubic in. 23) 52 square units.
E. 24) Area covers surfaces in square units; volume fills space in cubic units. 25) Triangle area= 42 square units.
EXIT. 26) height= 6 cm; $SA=312$ square cm.
Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.
F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may confuse lateral faces, base area, and volume. Sketch or use a net and label every face once.

TEACHER CHECK

- ☐ Uses an appropriate Grade 6 Math strategy.
- ☐ Shows or explains thinking clearly.
- ☐ Checks a response using evidence, a rule, or a second method.
- ☐ Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.